

COMPUTATIONAL GAME THEORY

Exercises

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ABOUT THIS DOCUMENT

The purpose of these exercises is to help you, as students of *Computational Game Theory*, actively engage with the concepts introduced in the lectures. You are encouraged to develop your own approaches to solving the problems and to use the exercises as an opportunity to deepen your understanding of the material. Most exercises can be solved with pen and paper, and you are welcome to use the wide margins of this document for notes, calculations, and sketches of arguments.

When needed, consult the online course materials for the relevant definitions, fundamental results, and examples. Many of the exercises are adapted from the recommended textbooks [1, 2, 3, 4]. Some questions are more demanding and require longer mathematical arguments; these are marked with ★.

In addition to exercises directly related to the lecture content, the first section recalls selected mathematical prerequisites needed for the study of game theory. This section is based primarily on material from undergraduate courses in linear algebra, optimization, linear programming, discrete mathematics, and probability theory.

1 MATHEMATICAL PREREQUISITES

Exercise 1.1

Duality in linear programming. The concept of duality appears in many game-theoretic problems. In this exercise we consider the following linear program:

$$\begin{array}{ll} \text{Minimize} & -x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & -x_1 + 4x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{array}$$

Characterize the feasible set of the problem by its vertices (extreme points). Additionally, find the optimal solution of this linear program and formulate the dual program. What is the meaning of the dual program and its relation to the original program? What will be the optimal value of the dual problem?

Exercise 1.2

Minimax/maximin values. Let X and Y be nonempty sets and consider a real function $f: X \times Y \rightarrow \mathbb{R}$. Show that

$$\max_{x \in X} \min_{y \in Y} f(x, y) \leq \min_{y \in Y} \max_{x \in X} f(x, y), \quad (1)$$

where we assume that all maxima and minima above exist. Find an example of function f taking at least two different values such that (a) the inequality is strict and (b) the inequality becomes an equality. *Hint:* Such examples exist already when both X and Y have 2 elements.

Exercise 1.3

Saddle points. Let $f: X \times Y \rightarrow \mathbb{R}$ be a function, where X and Y are arbitrary nonempty set. We say that $(x^*, y^*) \in X \times Y$ is a *saddle point* of f if

$$f(x, y^*) \leq f(x^*, y^*) \leq f(x^*, y) \quad \text{for all } x \in X, y \in Y. \quad (2)$$

Show that a function f has a saddle point if, and only if,

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y), \quad (3)$$

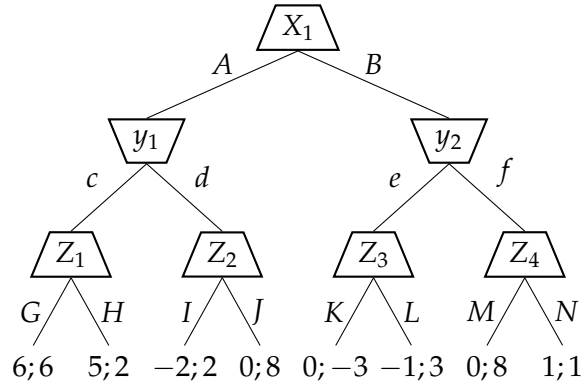
provided that all the maxima/minima above exist.

Exercise 1.4

Consider a game called Nim played by two players. There are 21 matches on board. The players take turns removing 1, 2, or 3 matches from the board. The player that takes the last match loses. What is the strategy you should play in such a game?

Exercise 1.5

Consider following sequential game played by 2 players. What is the rational strategy of both players in such a game, if player 1 acts first and if player 2 acts first.


Exercise 1.6

Let $N = \{1, \dots, n\}$ and fix a player $i \in N$. A coalition $A \subseteq N \setminus \{i\}$ is drawn at random. Compute the probability of obtaining a given coalition A under each of the following processes.

- A is drawn uniformly from all subsets of $N \setminus \{i\}$.
- A size a is drawn uniformly from $\{0, 1, \dots, n-1\}$, and then A is drawn uniformly from the subsets of $N \setminus \{i\}$ of size a .
- The n players are arranged in a uniformly random order, and A is the set of players preceding i .

SOLUTIONS

Solution 1.1

In any linear programming problem, the feasible set is a convex polyhedron. In our case the polyhedron is bounded, so it is a convex polygon characterized by its vertices. Every vertex corresponds to a unique solution of some linear equality system associated with a subset of linear inequality constraints of the problem. For example, a point $(1, 3)$ is a vertex since it is a unique solution to linear equations $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5$. The remaining vertices are $(0, \frac{1}{2})$, $(0, 4)$, $(2, 1)$. See Figure 1, which is an output of the simple online solver <https://online-optimizer.appspot.com>.

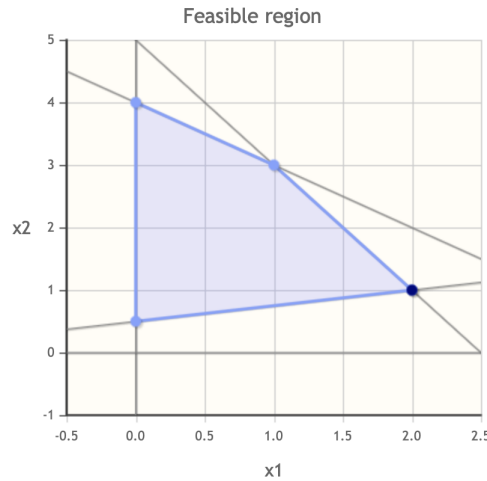


Figure 1: The feasible set corresponding to constraints (5)–(7)

What is the optimal solution to the problem? The fundamental theorem of linear programming says that, if a linear program has an optimal solution, then it is attained at some vertex of the feasible set. Clearly, at least one optimal solution exists in our case since the feasible set is bounded. Therefore, it suffices to compute the values of the objective function $f(x_1, x_2) = -x_1 + 2x_2$ for all the vertices and select the one with the minimal value. This yields the optimal solution $(2, 1)$ with the optimal value is $f(2, 1) = 0$.

To formulate the dual problem, it is convenient to formulate the original problem so that all inequalities are of the form $\geq$:

$$\text{Minimize} \quad -x_1 + 2x_2 \quad (4)$$

$$\text{subject to} \quad -x_1 - x_2 \geq -4 \quad (5)$$

$$-2x_1 - x_2 \geq -5 \quad (6)$$

$$-x_1 + 4x_2 \geq 2 \quad (7)$$

$$x_1, x_2 \geq 0 \quad (8)$$

By design, the dual program has 3 variables y_1, y_2, y_3 corresponding to the 3 primal linear constraints (5)–(7), and 2 linear constraints associated with

2 primal variables x_1, x_2 . The exact form of the dual program can be found in any textbook on linear programming. Specifically the dual is

$$\begin{aligned} \text{Maximize} \quad & -4y_1 - 5y_2 + 2y_3 \\ \text{subject to} \quad & -y_1 - 2y_2 - y_3 \leq -1 \\ & -y_1 - y_2 + 4y_3 \leq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

The duality theorem immediately implies that the primal and dual problems have the same optimal value 0.

What is the interpretation of duality? The optimal value of dual problem provides the tightest lower bound on the solution of the primal problem. In fact, the dual problem controls the optimal value of the primal problem from below, and does so in the best possible way — the optimal values of both programs coincide!

We will elaborate on the idea sketched above. How to obtain a lower bound for the primal problem? Note that one possible lower bound on the values of the objective f follows from the nonnegativity constraints (8) and the constraint (5) since

$$-4 \leq -x_1 - x_2 \leq -x_1 + 2x_2.$$

Can we do better? That is, can we obtain a higher value than -4 for the lower bound? It is easy to see that by multiplying the inequalities (5) and (7) by $\frac{1}{2}$ and adding them we obtain the bound

$$-1 \leq -x_1 + \frac{3}{2}x_2 \leq -x_1 + 2x_2,$$

which is tighter. Can we still improve on this lower bound? Let us try to generalize the idea of combining the linear inequalities of the primal problem for estimating the optimal value from below. The goal is to find nonnegative real numbers y_1, y_2, y_3 such that

$$\begin{aligned} -4y_1 - 5y_2 + 2y_3 &\leq y_1(-x_1 - x_2) + y_2(-2x_1 - x_2) + y_3(-x_1 + 4x_2) \\ &= (-y_1 - 2y_2 - y_3)x_1 + (-y_1 - y_2 + 4y_3)x_2 \\ &\leq -x_1 + 2x_2 \end{aligned}$$

and the second inequality should be as tight as possible. One way to guarantee this is to maximize the most left-hand side $-4y_1 - 5y_2 + 2y_3$ while preserving the constraints $-y_1 - 2y_2 - y_3 \leq -1$ and $-y_1 - y_2 + 4y_3 \leq 2$ expressed by the second inequality above. In other words, we have derived precisely the dual problem.

Solution 1.2

To prove the inequality (1), define

$$F(x) = \min_{y \in Y} f(x, y), \quad x \in X,$$

and let $x^* \in X$ be the maximizer of F over X . Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \max_{x \in X} F(x) = F(x^*) = \min_{y \in Y} f(x^*, y).$$

For every $y \in Y$, we have

$$f(x^*, y) \leq \max_{x \in X} f(x, y)$$

by the definition of maximum, which implies

$$\min_{y \in Y} f(x^*, y) \leq \min_{y \in Y} \max_{x \in X} f(x, y).$$

Hence, (1) is proved.

We will construct two examples (a)-(b) under the assumption that $X = Y = \{1, 2\}$. First, consider a function f given by the matrix

		Y	
		1	2
X	1	0	1
	2	1	0

Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = 0 < 1 = \min_{y \in Y} \max_{x \in X} f(x, y).$$

For the second example (b), take a matrix

		Y	
		1	2
X	1	3	0
	2	2	2

Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = 2 = \min_{y \in Y} \max_{x \in X} f(x, y).$$

Solution 1.3

We define two functions: $F(x) = \min_{y \in Y} f(x, y)$, for all $x \in X$, and $G(y) = \max_{x \in X} f(x, y)$, for every $y \in Y$. Assume that f has a saddle point (x^*, y^*) . Then the saddle point definition (2) and the definition of maxima/minima imply

$$\begin{aligned} \min_{y \in Y} G(y) &\leq G(y^*) = \max_{x \in X} f(x, y^*) = f(x^*, y^*) \\ &= \min_{y \in Y} f(x^*, y) = F(x^*) \leq \max_{x \in X} F(x). \end{aligned}$$

We obtained the inequality

$$\min_{y \in Y} \max_{x \in X} f(x, y) \leq \max_{x \in X} \min_{y \in Y} f(x, y),$$

and since the converse inequality always holds (Exercise 2), we proved the minimax equality (3).

To prove the converse implication, assume that (3) holds. Let x^* and y^* be such that $F(x^*) = \max_{x \in X} F(x)$ and $G(y^*) = \min_{y \in Y} G(y)$, respectively. Then the assumption (3) gives

$$\begin{aligned} F(x^*) &= \max_{x \in X} F(x) = \max_{x \in X} \min_{y \in Y} f(x, y) \\ &= \min_{y \in Y} \max_{x \in X} f(x, y) = \min_{y \in Y} G(y) = G(y^*). \end{aligned} \tag{9}$$

Further, from the definition of F, G and x^*, y^* we get

$$F(x^*) \leq f(x^*, y^*) \leq G(y^*)$$

and (9) implies that even $F(x^*) = f(x^*, y^*) = G(y^*)$. Let $x \in X$ and $y \in Y$. Then

$$f(x, y^*) \leq G(y^*) = f(x^*, y^*) = F(x^*) \leq f(x^*, y).$$

Solution 1.4

Instead of asking what you should do when there are 21 matches, let us start with an easier version of this problem: There is only 1 match. In that case, the first player takes it and immediately loses, so the winner is player 2 in such a game. When there are 2 matches, the optimal strategy is to remove a single match and force player 2 to take the last one. The same logic can be applied for 3 and 4 matches. In those cases, the player 1 is the winner. Now, when there are 5 matches, player 1 is forced to take 1, 2 or 3 matches, which results in 2, 3, or 4 matches on board, respectively. The player is destined to lose in each of those if the opponent plays perfectly. So, no matter how many matches it takes, it loses.

Now, let us generalize this idea. There are a certain number of matches with which the player always loses against an optimal opponent. Let us call those losing amounts. We have seen that those amounts are 1 or 5. On the other hand, there are amounts from which the player can remove a certain amount of matches to end up with a losing amount for the opponent. We have seen that those are 2, 3 and 4. Let us call them winning amounts.

Since we know 5 is a losing amount from the player's perspective, then it is rational from the player's perspective to force the opponent to this losing amount. This can be achieved from 6, 7, 8 amounts. However, with 9 matches, the player cannot force the opponent into this losing state. Similar logic can be applied further ending up with losing amounts being

$$4n + 1 \quad \forall n \in \mathcal{N}$$

All the others are winning amounts because, from those, you will force the opponent into the losing amount.

The optimal strategy when there is a losing amount is to take an arbitrary amount of matches since you cannot win against a rational opponent. With the winning amount, the optimal strategy is always to take set amount of matches to end up in the losing amount.

Consider an algorithm called backward induction. This algorithm follows a similar approach as the one describes above. it starts at the end of the game tree (terminal states) and based on actions it goes backwards to the root (initial state) of the game.

Solution 1.5

Performing backward induction finds rational strategies. If player 1 is root, it also acts in Z_i . Since rational player is trying to maximize then in Z_1 , it will choose action G , because $6 > 5$. Similarly in Z_2 it will choose J because $-2 < 0$. Now the values of Z_1 and Z_2 are 6; 6 and 0; 8 respectively. So in y_1 , the rational player picks d , because $6 < 8$, which yields value of y_1 to 0; 8.

Similarly in the right subtree the rational player chooses K or N . This yields values for Z_3 and Z_4 to be $0; -3$ and $1; 1$ respectively. In y_2 it is then rational to choose f , so the value is $1; 1$. Lastly in X_1 the optimal action is B , because $0 < 1$. If both players play rationally, they will both receive reward 1.

When swapping the player the procedure is similar. Now in last decision nodes the player plays G, J, K, M , so the values are

$$Z_1 \rightarrow 6; 6 \quad (10)$$

$$Z_2 \rightarrow 0; 8 \quad (11)$$

$$Z_3 \rightarrow -1; 3 \quad (12)$$

$$Z_4 \rightarrow 0; 8 \quad (13)$$

The rational response for player 1 is to pick c and f

$$y_1 \rightarrow 6; 6 \quad (14)$$

$$y_2 \rightarrow 0; 8 \quad (15)$$

In the end the second player chooses action B because $6 < 8$. So the resulting reward would be 0 for player 1 and 8 for player 2.

Solution 1.6

(a) There are 2^{n-1} coalitions A to which player i doesn't belong, $i \notin A$. Therefore $p(A) = 2^{1-n}$ is the probability of picking such a coalition randomly. Consider now the second variant of random choice of the coalition. Let A be a random coalition selected in this way. We can compute the probability $q(A)$ using the law of total probability,

(b) The process produces A precisely when it draws the size $|A|$, which happens with probability $1/n$, and then selects A among the $\binom{n-1}{|A|}$ coalitions of that size, which happens with probability $1/\binom{n-1}{|A|}$. Hence

$$q(A) = \frac{1}{n} \cdot \frac{1}{\binom{n-1}{|A|}}.$$

This is a probability distribution:

$$\sum_{A \subseteq N \setminus \{i\}} q(A) = \frac{1}{n} \sum_{a=0}^{n-1} \binom{n-1}{a} \frac{1}{\binom{n-1}{a}} = \frac{1}{n} \cdot n = 1.$$

(b) = (c). Expanding the binomial coefficient in (b) and using $n(n-1)! = n!$,

$$\frac{1}{n \binom{n-1}{a}} = \frac{1}{n} \cdot \frac{a!(n-1-a)!}{(n-1)!} = \frac{a!(n-1-a)!}{n!},$$

which is exactly the expression obtained in (c). The two mechanisms therefore induce the same distribution. This is the weighting that appears in the Shapley value; the uniform distribution p of (a) is the one appearing in the Banzhaf value. Note that both p and q depend only on the size of each coalition. Specifically, if A and B are coalitions such that $|A| = |B|$, then $p(A) = p(B) = 2^{1-n}$ and $q(A) = q(B)$. Probability distributions p and q appear in the Banzhaf and Shapley values in coalitional game theory. Observe that there is a several order magnitude difference between them (Figure 2). For example, in case that there are $n = 60$ players, the probabilities

of coalition A with $|A| = 10$ are $p(A) \approx 1.7 \times 10^{-18}$ and $q(A) \approx 2.6 \times 10^{-13}$, respectively.

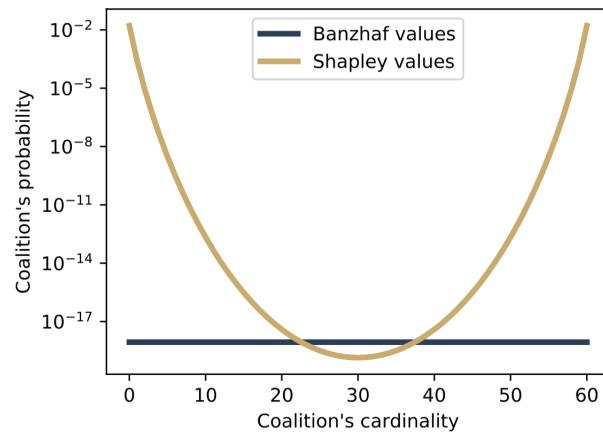


Figure 2: Values of p and q .

2 NORMAL-FORM GAMES

Exercise 2.7

Single attacker is about to attack one of the 4 military bases called T_1 , T_2 , T_3 , and T_4 . However, due to the budget restriction, the military can defend only one of the bases. If the attacker attacks the defended base, it gets a utility of -1 , while the military gets 2. On the other hand, when the attack is successful, the military does not receive anything, but the attacker gets the following rewards based on the base it attacked $T_1 \rightarrow 3$, $T_2 \rightarrow 7$, $T_3 \rightarrow 1$, and $T_4 \rightarrow 5$. (a) Formalize this situation as a normal-form game. (b) Let us assume that when the attacker attacks base, the alarm is triggered. If the attacked base is near the protected one, the military may dispatch striking team, which has 50 % chance to interrupt the attack. For base T_i , the nearby bases are T_{i-1} and T_{i+1} . Bases T_1 and T_4 are also close to each other. (c) Find all pure Nash equilibria in games (a) and (b).

Exercise 2.8

Is there a game with at least one pure Nash equilibrium which is removed during the process of iterated elimination? Either show that such a game cannot exist or find an example that it does.

Exercise 2.9

Consider a two-player normal-form game with the payoff matrix

		Player 2		
		V	S	R
Player 1	U	1, 3	4, 2	-1, 2
	C	1, 0	2, -2	0, -1
	D	1, 2	-1, 1	3, 3

(a) Find all Pareto optimal outcomes. (b) Find all pure Nash equilibria. (c) Find all dominated pure strategies and apply the procedure of iterative removal of dominated strategies to this game.

Exercise 2.10

We present the game called *Battle of the Sexes*. Its name is derived from the situation where a couple (Alice and Bob) is trying to plan what to do on Saturday. The alternatives are going to a concert (C) or watching a football match (F). Bob prefers football and Alice prefers the concert, but both prefer being together to being alone, even if that means agreeing to the less-preferred recreational activity.

		Bob	
		C	F
Alice	C	2, 1	0, 0
	F	0, 0	1, 2

Find equilibrium strategies of this game.

Exercise 2.11

We present the game called *Game of chicken*. Two drivers race towards each other on a narrow road. Each may *swerve* (S) or *drive straight* (D). A driver who swerves while the other drives straight loses face; a driver who drives straight while the other swerves wins the contest. If both swerve, nothing happens; if neither swerves, they crash.

		Driver 2	
		S	D
Driver 1	S	0, 0	-1, 1
	D	1, -1	-10, -10

- (a) Find all Nash equilibrium strategies of this game.
- (b) Driver 1 is riding a motorcycle and Driver 2 an armoured car, so a crash costs them -10 and -5 respectively. Before computing: which driver do you expect to drive straight more often? Now find the mixed equilibrium.
- (★) (c) Let the crash cost driver i the amount $-c_i$, where $c_i > 1$. Find the mixed equilibrium and the equilibrium payoffs.

SOLUTIONS

Solution 2.7

(a) We use the military as a row player and the attacker as a column player. The first value corresponds to the utility of the defender, while the second corresponds to the attacker's utility. The utility matrix in the normal-form game is then

		Attacker			
		T ₁	T ₂	T ₃	T ₄
Military	T ₁	2, -1	0, 7	0, 1	0, 5
	T ₂	0, 3	2, -1	0, 1	0, 5
	T ₃	0, 3	0, 7	2, -1	0, 5
	T ₄	0, 3	0, 7	0, 1	2, -1

(b) In the modified game, when the attacker attacks the defended base, the result is still the same. But when the base is nearby the protected base, the utility changes in the following way.

$$u_1(\mathbf{T}_i, \mathbf{T}_{i+1}) = \frac{1}{2}u_1(\mathbf{T}_{i+1}, \mathbf{T}_{i+1}) + \frac{1}{2} \cdot 2$$

$$u_2(\mathbf{T}_i, \mathbf{T}_{i+1}) = \frac{1}{2}u_2(\mathbf{T}_{i+1}, \mathbf{T}_{i+1}) + \frac{1}{2} \cdot (-1)$$

The utility matrix in the modified game is

		Attacker			
		T ₁	T ₂	T ₃	T ₄
Military	T ₁	2, -1	1, 3	0, 1	1, 2
	T ₂	1, 1	2, -1	1, 0	0, 5
	T ₃	0, 3	1, 3	2, -1	1, 2
	T ₄	1, 1	0, 7	1, 0	2, -1

(c) Neither game has any pure Nash equilibrium.

Solution 2.8

In order to remove a strategy during the iterated elimination, it has to be strictly dominated. Now, let us suppose that there exists a Nash equilibrium $\mathbf{s}^* \in \mathbf{S}$ in which player i plays strategy $s_i^* \in S_i$ that is strictly dominated by $s_i \in S_i$, that is,

$$u_i(s_i, \mathbf{s}_{-i}^*) > u_i(s_i^*, \mathbf{s}_{-i}^*) = u_i(\mathbf{s}^*).$$

This is in direct contradiction with the definition of a Nash equilibrium. Hence, the iterated elimination of strictly dominated strategies cannot remove any Nash equilibrium.

On the other hand, let us suppose that there exist a Nash equilibrium $\mathbf{s}^* \in \mathbf{S}$ in which player i plays strategy $s_i^* \in S_i$ that is only weakly dominated by $s_i \in S_i$ such that for some $\mathbf{s}_{-i}' \in \mathbf{S}_{-i}$ the following holds:

$$u_i(s_i, \mathbf{s}_{-i}') > u_i(s_i^*, \mathbf{s}_{-i}').$$

However, since $\mathbf{s}_{-i}' = \mathbf{s}_{-i}^*$ is not true in general, the above inequality does not necessarily violate the Nash equilibrium condition. This observation is instrumental in finding a counterexample with $\mathbf{s}_{-i}' \neq \mathbf{s}_{-i}^*$. For example:

		Player 2	
		d	e
Player 1	a	0, 1	1, 0
	b	-1, 0	1, 2
	c	1, 0	0, 1

Solution 2.9

- (a) The only Pareto optimal strategies are $(\mathbf{D}, \mathbf{R})$ and $(\mathbf{U}, \mathbf{S})$.
 (b) Pure Nash equilibria are $(\mathbf{D}, \mathbf{R})$, $(\mathbf{U}, \mathbf{V})$, $(\mathbf{C}, \mathbf{V})$.
 (c) Pure strategy $\mathbf{V}$ strictly dominates $\mathbf{S}$ (also $\mathbf{R}$ weakly dominates $\mathbf{S}$).

		Player 2	
		$\mathbf{V}$	$\mathbf{R}$
Player 1	$\mathbf{U}$	1, 3	-1, 2
	$\mathbf{C}$	1, 0	0, -1
	$\mathbf{D}$	1, 2	3, 3

After removing $\mathbf{S}$, both $\mathbf{U}$ and $\mathbf{C}$ are weakly dominated by $\mathbf{D}$.

		Player 2	
		$\mathbf{V}$	$\mathbf{R}$
Player 1	$\mathbf{D}$	1, 2	3, 3

Now, $\mathbf{R}$ strictly dominates $\mathbf{V}$. This leaves us with only one strategy profile $(\mathbf{D}, \mathbf{R})$, which is also a Nash equilibrium.

Solution 2.10

It is easy to verify that both strategy profiles (C, C) and (F, F) are equilibria in pure strategies. We will show that the game has a mixed strategy equilibrium, too.

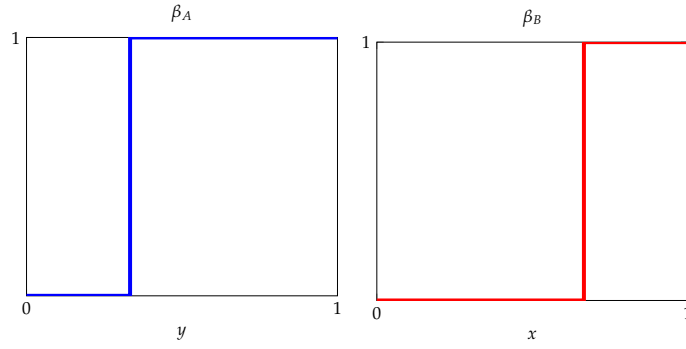
Let $x \in [0, 1]$ and $y \in [0, 1]$ be the probabilities of playing C for Alice and Bob, respectively. Since each player has only two pure strategies, the set of all mixed strategies can be viewed as the unit interval $[0, 1]$. Thus, the expected utility of Alice is $U_A(x, y) = 2xy + (1 - x)(1 - y)$ and the expected utility of Bob is $U_B(x, y) = xy + 2(1 - x)(1 - y)$, for all $x, y \in [0, 1]$.

We compute the best responses of both players. For Alice this is the mapping defined by

$$\beta_A(y) = \arg \max_{x \in [0, 1]} U_A(x, y), \quad y \in [0, 1],$$

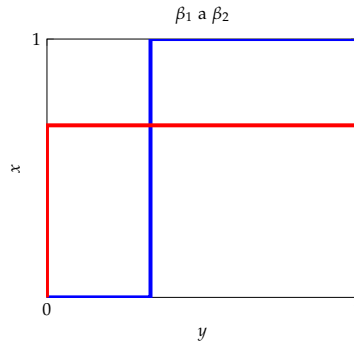
and analogously for Bob. We get

$$\beta_A(y) = \begin{cases} 0 & 0 \leq y < \frac{1}{3}, \\ [0, 1] & y = \frac{1}{3}, \\ 1 & \frac{1}{3} < y \leq 1, \end{cases} \quad \beta_B(x) = \begin{cases} 0 & 0 \leq x < \frac{2}{3}, \\ [0, 1] & x = \frac{2}{3}, \\ 1 & \frac{2}{3} < x \leq 1. \end{cases}$$



We know that $(x^*, y^*) \in [0, 1]^2$ would correspond to an equilibrium in mixed strategies if, and only if,

$$x^* \in \beta_A(y^*) \text{ and } y^* \in \beta_B(x^*). \quad (16)$$



The geometric interpretation of the condition (17) is that $(x^*, y^*) \in [0, 1]^2$ is the point of the common intersection of the graphs of β_A and β_B . Therefore we obtain a mixed strategy equilibrium in which Alice plays the mixed strategy $(\frac{2}{3}, \frac{1}{3})$ and Bob uses the mixed strategy $(\frac{1}{3}, \frac{2}{3})$.

Solution 2.11

(A) It is easy to verify that both strategy profiles (S, D) and (D, S) are equilibria in pure strategies. We will show that the game has a mixed strategy equilibrium, too.

Let $p \in [0, 1]$ and $q \in [0, 1]$ be the probabilities of swerving for Driver 1 and Driver 2. Since each player has only two pure strategies, the set of all mixed strategies can be viewed as the unit interval $[0, 1]$. Thus, the expected utility of Driver 1 is $U_1(p, q) = 9p + 11q - 10pq - 10$ and the expected utility of Driver 2 is $U_2(p, q) = U_1(q, p)$, for all $p, q \in [0, 1]$.

We compute the best responses of both players. For Driver 1 this is the mapping defined by

$$\beta_1(q) = \arg \max_{p \in [0, 1]} U_1(p, q), \quad q \in [0, 1],$$

and analogously for Driver 2. We get

$$\beta_1(q) = \begin{cases} 1 & 0 \leq q < \frac{9}{10}, \\ [0, 1] & q = \frac{9}{10}, \\ 0 & \frac{9}{10} < q \leq 1, \end{cases} \quad \beta_2(p) = \begin{cases} 1 & 0 \leq p < \frac{9}{10}, \\ [0, 1] & p = \frac{9}{10}, \\ 0 & \frac{9}{10} < p \leq 1. \end{cases}$$

A pair $(p^*, q^*) \in [0, 1]^2$ is an equilibrium in mixed strategies if, and only if,

$$p^* \in \beta_1(q^*) \quad \text{and} \quad q^* \in \beta_2(p^*).$$

The geometric interpretation of this condition is that (p^*, q^*) is a point of the common intersection of the graphs of β_1 and β_2 . Besides the two pure equilibria $(1, 0)$ and $(0, 1)$, we therefore obtain a mixed strategy equilibrium in which each driver swerves with probability $\frac{9}{10}$. The payoffs are $-\frac{1}{10}$ for both drivers, although both prefer the outcome (S, S) with payoffs $(0, 0)$.

(B) Repeating the computation with the crash payoffs -10 and -5 , the best responses of Driver 1 and Driver 2 change their value at $q = \frac{9}{10}$ and $p = \frac{4}{5}$, respectively. Hence Driver 1 drives straight with probability $\frac{1}{5}$ and Driver 2 with probability $\frac{1}{10}$, and the payoffs are $-\frac{1}{10}$ and $-\frac{1}{5}$. The driver who fears the crash more is the one who drives straight more often, since each driver's equilibrium probability is the point at which the *opponent's* best response changes, and hence depends only on the opponent's payoffs.

(*) (c) In general $U_1(p, q) = -p + q - c_1(1 - p)(1 - q)$, and the best response of Driver 1 changes its value at $q^* = 1 - \frac{1}{c_1}$; symmetrically, that of Driver 2 at $p^* = 1 - \frac{1}{c_2}$. Substituting back gives $U_1(p^*, q^*) = -\frac{1}{c_1}$ and $U_2(p^*, q^*) = -\frac{1}{c_2}$. Thus each driver drives straight with probability $\frac{1}{c_j}$, where c_j is the crash payoff magnitude of the *opponent*, and receives $-\frac{1}{c_i}$.

3 COMPUTING NASH EQUILIBRIA

Exercise 3.12

Battle of the Sexes. Alice (rows) and Bob (columns) choose Concert (C) or Football (F).

	C	F
C	(2, 1)	(0, 0)
F	(0, 0)	(1, 2)

Let $x = P(\text{Alice plays C})$ and $y = P(\text{Bob plays C})$.

- Write Alice's expected payoff as a function of x and y . Draw both best-response correspondences in the unit square and read off all Nash equilibria.
- Find all Nash equilibria again by support enumeration: go through all 9 pairs of supports. Write the pair $(\{C, F\}, \{C, F\})$ as a linear feasibility problem.

Exercise 3.13

Chicken. Two drivers head towards each other and choose Swerve (S) or Straight (D).

	S	D
S	(0, 0)	(-1, 1)
D	(1, -1)	(-10, -10)

Let $p = P(\text{driver 1 swerves})$ and $q = P(\text{driver 2 swerves})$.

- Write U_1 as a function of p and q . Draw both best-response correspondences and read off all Nash equilibria.
- In Battle of the Sexes, Alice's best response steps *up* as y grows; in Chicken, driver 1's best response steps *down* as q grows. Why?
- (★) Replace the crash payoff -10 by $-c$ with $c > 1$. In the mixed equilibrium, how often does each driver go straight, and what does each get?

Exercise 3.14

Patrol chase. The military defends one of four bases $T_1, \dots, T_4$; at the same time the attacker attacks one base. If the attacked base is defended, the military gets 2 and the attacker -1 . Otherwise the military gets 0 and the attacker gets the value v_j of the base, where

$$(v_1, v_2, v_3, v_4) = (3, 7, 1, 5).$$

Let d_j be the probability that the military defends T_j and a_j the probability that the attacker attacks T_j .

- Show that $U_M(T_j) = 2a_j$ and $U_A(T_j) = v_j - d_j(v_j + 1)$.
- Guess that both players use the same support T . Show that the military's indifference forces the attacker to attack uniformly on T .

- (c) Show that every base in T is worth at least as much as every base outside T . Which candidate sets T remain?
- (d) Test the candidates and find the Nash equilibrium. Which base is never defended, and which is defended most?

Hint for (c): inside T the attacker earns $e_A = v_j - d_j(v_j + 1) \leq v_j$; outside T he would earn v_j and this must not beat e_A .

Exercise 3.15

A degenerate game.

	L	R
T	(1, 1)	(0, 0)
B	(1, 0)	(2, 1)

- (a) Find all Nash equilibria.
- (b) Which of them would support enumeration miss if it tested only supports of equal size? Explain by counting equations.

Exercise 3.16

Quantal response in asymmetric Matching Pennies. Alice (rows) wants to match Bob (columns), Bob wants to mismatch. Alice's payoff at (T, L) is 9 instead of 1.

	L	R
T	(9, 0)	(0, 1)
B	(0, 1)	(1, 0)

Let $x = P(\text{Alice plays } T)$ and $y = P(\text{Bob plays } L)$. The logit quantal response of player i with precision $\lambda \geq 0$ plays s_i with probability

$$\text{QR}_i^\lambda(p_{-i})(s_i) = \frac{\exp(\lambda U_i(s_i, p_{-i}))}{\sum_{t_i} \exp(\lambda U_i(t_i, p_{-i}))},$$

and a profile p is a quantal response equilibrium (QRE) if $p = \text{QR}^\lambda(p)$.

- (a) Find the Nash equilibrium. How does Alice's payoff 9 influence her own equilibrium strategy?
- (b) Show that Alice's quantal response to y is

$$P(T) = \frac{1}{1 + e^{-\lambda d}}, \quad d = U_A(T) - U_A(B) = 10y - 1.$$

For $\lambda = 1$, compute it for $y = 0, 0.1, 0.3$ and compare with Alice's best response.

- (c) Let $\lambda = 1$. Which of the following profiles is a QRE?

$$\text{A: } (x, y) = (0.8, 0.3), \quad \text{B: } (x, y) = (0.5, 0.1), \quad \text{C: } (x, y) = (0.894, 0.313).$$

- (d) Compare the Nash equilibrium with the QRE. Why is the Nash equilibrium not a QRE?

SOLUTIONS

Solution 3.12

(a) It is easy to verify that both strategy profiles (C, C) and (F, F) are equilibria in pure strategies. We will show that the game has a mixed strategy equilibrium, too.

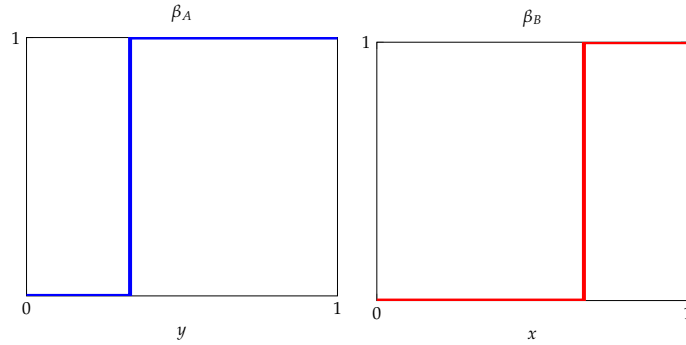
Let $x \in [0, 1]$ and $y \in [0, 1]$ be the probabilities of playing C for Alice and Bob, respectively. Since each player has only two pure strategies, the set of all mixed strategies can be viewed as the unit interval $[0, 1]$. Thus, the expected utility of Alice is $U_A(x, y) = 2xy + (1 - x)(1 - y)$ and the expected utility of Bob is $U_B(x, y) = xy + 2(1 - x)(1 - y)$, for all $x, y \in [0, 1]$.

We compute the best responses of both players. For Alice this is the mapping defined by

$$\beta_A(y) = \arg \max_{x \in [0, 1]} U_A(x, y), \quad y \in [0, 1],$$

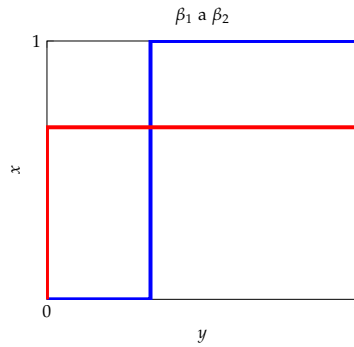
and analogously for Bob. We get

$$\beta_A(y) = \begin{cases} 0 & 0 \leq y < \frac{1}{3}, \\ [0, 1] & y = \frac{1}{3}, \\ 1 & \frac{1}{3} < y \leq 1, \end{cases} \quad \beta_B(x) = \begin{cases} 0 & 0 \leq x < \frac{2}{3}, \\ [0, 1] & x = \frac{2}{3}, \\ 1 & \frac{2}{3} < x \leq 1. \end{cases}$$



We know that $(x^*, y^*) \in [0, 1]^2$ would correspond to an equilibrium in mixed strategies if, and only if,

$$x^* \in \beta_A(y^*) \text{ and } y^* \in \beta_B(x^*). \quad (17)$$



The geometric interpretation of the condition (17) is that $(x^*, y^*) \in [0, 1]^2$ is the point of the common intersection of the graphs of β_A and β_B . Therefore we

obtain a mixed strategy equilibrium in which Alice plays the mixed strategy $(\frac{2}{3}, \frac{1}{3})$ and Bob uses the mixed strategy $(\frac{1}{3}, \frac{2}{3})$.

(b) The nine pairs fall into four groups.

- $(\{C\}, \{C\})$ and $(\{F\}, \{F\})$: no one gains by deviating ($2 \geq 0$, $1 \geq 0$; $1 \geq 0$, $2 \geq 0$). Equilibria.
- $(\{C\}, \{F\})$ and $(\{F\}, \{C\})$: Alice gets 0 but the other row gives 1 or 2. The outside inequality fails.
- One pure, one mixed, e.g. $(\{C\}, \{C, F\})$: Bob must be indifferent against C, i.e. $1 = 0$. The equalities contradict each other. The same happens for the other three such pairs ($0 = 2$, $2 = 0$, $0 = 1$).
- $(\{C, F\}, \{C, F\})$: find x, y, e_1, e_2 with

$$2y - e_1 = 0, \quad (1 - y) - e_1 = 0, \quad x - e_2 = 0, \quad 2(1 - x) - e_2 = 0, \quad 0 \leq x, y \leq 1.$$

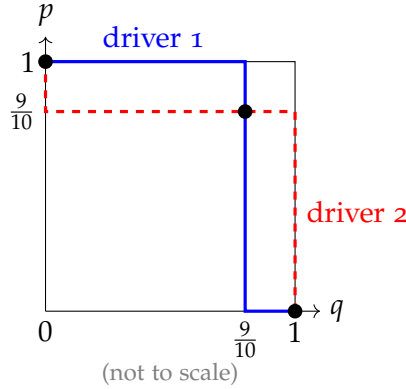
This is an LP with a constant objective. Solution $y = \frac{1}{3}$, $x = \frac{2}{3}$, $e_1 = e_2 = \frac{2}{3}$. No outside inequalities, since both supports are full.

Solution 3.13

(a) $U_1(S) = q \cdot 0 + (1 - q)(-1) = q - 1$ and $U_1(D) = q - 10(1 - q) = 11q - 10$, so

$$U_1(p, q) = p(q - 1) + (1 - p)(11q - 10) = p(9 - 10q) + 11q - 10.$$

Driver 1 swerves for $q < \frac{9}{10}$, goes straight for $q > \frac{9}{10}$, anything at $q = \frac{9}{10}$. Driver 2 is symmetric.



Three equilibria: (S, D) , (D, S) , and both swerve with probability $\frac{9}{10}$, each getting $q - 1 = -\frac{1}{10}$.

(b) In BoS Alice wants to *match* Bob: the more Bob goes to the concert, the better C is for her (coefficient $3y - 1$ grows in y). In Chicken a driver wants to *do the opposite*: the more the other swerves, the better straight is (coefficient $9 - 10q$ falls in q).

(★c) $U_1(S) = q - 1$, $U_1(D) = q - c(1 - q)$. Equal iff $q = 1 - \frac{1}{c}$. So each driver goes straight with probability $\frac{1}{c}$ and gets $-\frac{1}{c}$. A more dangerous crash makes each driver go straight less often, and the expected loss shrinks.

Solution 3.14

(a) Defending T_j pays 2 exactly when T_j is attacked: $U_M(T_j) = 2a_j$. Attacking T_j pays -1 with probability d_j and v_j otherwise: $U_A(T_j) = -d_j + (1 - d_j)v_j = v_j - d_j(v_j + 1)$.

(b) The military is indifferent on T : $2a_j = e_M$ for all $j \in T$. So all a_j on T are equal: $a_j = \frac{1}{|T|}$ and $e_M = \frac{2}{|T|}$. The attacker's mix is fixed by the *military's* payoffs; the values v_j do not enter.

(c) Inside T : $e_A = v_j - d_j(v_j + 1) \leq v_j$. Outside T : $d_j = 0$, so attacking pays v_j , and we need $v_j \leq e_A$. Hence every value inside is $\geq e_A \geq$ every value outside: T consists of the most valuable bases. Ordered by value $T_2(7)$, $T_4(5)$, $T_1(3)$, $T_3(1)$, the candidates are

$$\{T_2\}, \quad \{T_2, T_4\}, \quad \{T_1, T_2, T_4\}, \quad \{T_1, T_2, T_3, T_4\}.$$

(d) The attacker's indifference gives $d_j = \frac{v_j - e_A}{v_j + 1}$ on T , and $\sum_{j \in T} d_j = 1$ fixes e_A .

- $\{T_2\}$: $d_2 = 1$, $e_A = 7 - 8 = -1$. Attacking T_4 pays $5 > -1$. Rejected.
- $\{T_2, T_4\}$: $a_2 = a_4 = \frac{1}{2}$. From $\frac{7-e_A}{8} + \frac{5-e_A}{6} = 1$: $e_A = \frac{17}{7}$, $d = (\frac{4}{7}, \frac{3}{7})$. Attacking T_1 pays $3 > \frac{17}{7}$. Rejected.
- $\{T_1, T_2, T_4\}$: $a_1 = a_2 = a_4 = \frac{1}{3}$. From $\frac{3-e_A}{4} + \frac{7-e_A}{8} + \frac{5-e_A}{6} = 1$: $e_A = \frac{35}{13}$, so

$$(d_1, d_2, d_4) = (\frac{1}{13}, \frac{7}{13}, \frac{5}{13}).$$

Outside: attacking T_3 pays $1 \leq \frac{35}{13}$; defending T_3 pays $0 \leq \frac{2}{3} = e_M$.
Equilibrium.

- All four: $e_A = \frac{47}{25}$ and $d_3 = \frac{1-47/25}{2} = -\frac{11}{25} < 0$. Rejected.

Nash equilibrium: the military defends T_1, T_2, T_4 with probabilities $\frac{1}{13}, \frac{7}{13}, \frac{5}{13}$; the attacker attacks T_1, T_2, T_4 with probability $\frac{1}{3}$ each. T_3 is never defended (and never attacked); T_2 is defended most. Payoffs: military $\frac{2}{3}$, attacker $\frac{35}{13}$.

Every step was an indifference condition, except the rejections: $\{T_2\}$ and $\{T_2, T_4\}$ fail the outside inequality, the full set fails non-negativity.

Solution 3.15

(a) Let $x = P(T)$, $y = P(L)$. Alice gets $U_A(T) = y$ and $U_A(B) = y + 2(1 - y) = 2 - y$, so B is strictly better unless $y = 1$, where she is indifferent. Bob gets $U_B(L) = x$ and $U_B(R) = 1 - x$.

- If $y < 1$, Alice plays B , so Bob plays R : the equilibrium (B, R) .
- If $y = 1$, any x is a best response for Alice, and L is a best response for Bob iff $x \geq 1 - x$, i.e. $x \geq \frac{1}{2}$.

Nash equilibria: (B, R) and the whole segment $\{(xT + (1 - x)B, L) : \frac{1}{2} \leq x \leq 1\}$, infinitely many.

(b) The segment uses supports $\{T, B\}$ and $\{L\}$ (for $x < 1$), of different sizes. Alice's mix has one free number; it would be pinned by Bob's indifference, but Bob uses a single strategy, so there is no equation. Only the

inequality $U_B(R) \leq U_B(L)$ bounds x , which gives an interval. Equal-size enumeration finds only (T, L) and (B, R) .

The cause is a tie: Bob's pure strategy L (one strategy) has two best responses for Alice, T and B . A game where some mixed strategy with k strategies in its support has more than k pure best responses is called *degenerate*.

Solution 3.16

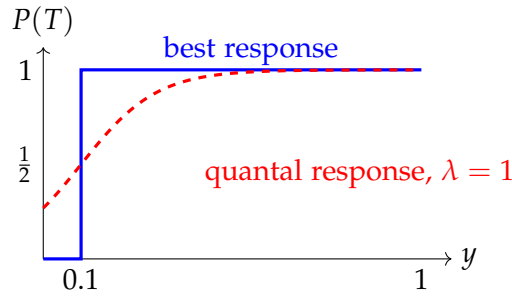
(a) Alice is indifferent iff $U_A(T) = 9y$ equals $U_A(B) = 1 - y$, i.e. $y = \frac{1}{10}$. Bob is indifferent iff $U_B(L) = 1 - x$ equals $U_B(R) = x$, i.e. $x = \frac{1}{2}$. The unique Nash equilibrium is $x = \frac{1}{2}$, $y = \frac{1}{10}$. Alice's payoff 9 changes only Bob's strategy: Alice still plays T with probability $\frac{1}{2}$, exactly as in ordinary Matching Pennies, because her mix is pinned by Bob's indifference.

(b) Dividing the whole fraction by $e^{\lambda U_A(T)}$,

$$p(T) = \frac{e^{\lambda \cdot 9y}}{e^{\lambda \cdot 9y} + e^{\lambda(1-y)}} = \frac{1}{1 + e^{-\lambda(10y-1)}}.$$

Only the payoff difference $d_A = 10y - 1$ matters. For $\lambda = 1$:

y	d_A	best response $P(T)$	quantal response $P(T)$
0	-1	0	0.27
0.1	0	any $x \in [0, 1]$	0.5
0.3	2	1	0.88



At $y = 0.1$ the best response is any mix, the quantal response is exactly $\frac{1}{2}$. For $\lambda = 0$ the curve is flat at $\frac{1}{2}$; for $\lambda \rightarrow \infty$ it becomes the step.

(c) Check both players: Alice's quantal response to y must return x (with $d_A = 10y - 1$), and Bob's quantal response to x must return y (with $d_B = U_B(L) - U_B(R) = 1 - 2x$).

	Alice: $d_A \rightarrow P(T)$	Bob: $d_B \rightarrow P(L)$	
A	$2 \rightarrow 0.881 \neq 0.8$	$-0.6 \rightarrow 0.354 \neq 0.3$	not a QRE
B	$0 \rightarrow 0.5 = 0.5 \checkmark$	$0 \rightarrow 0.5 \neq 0.1$	not a QRE
C	$2.13 \rightarrow 0.894 \checkmark$	$-0.788 \rightarrow 0.313 \checkmark$	QRE

(d) The two predictions differ considerably:

	Nash	QRE ($\lambda = 1$)
$x = P(T)$	0.5	0.894
$y = P(L)$	0.1	0.313

In the QRE Alice's own payoff 9 matters: she plays T much more often. Bob anticipates this and mostly plays R , but being noisy he still plays L with

probability 0.313. As λ grows, the QRE approaches the Nash equilibrium: (0.702, 0.117) for $\lambda = 5$ and (0.522, 0.100) for $\lambda = 50$.

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